

Writing Small Numbers in Scientific NotationQuestion Bank - Additional 3.09 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 4, write each number in scientific notation.**

1. 0.008

$$8 \times 10^{-3}$$

2. 0.00000026

$$2.6 \times 10^{-7}$$

3. 0.0475

$$4.75 \times 10^{-2}$$

4. 0.000000000091 seconds

$$9.1 \times 10^{-11} \text{ s}$$

For questions 5 and 6, write each number in standard form.

- 5.
- 7×10^{-5}

$$0.00007$$

- 6.
- 2.6×10^{-8}

$$0.000000026$$

Powers of ten.

7. Write each value as a single power of 10, then as a fraction or decimal.

- a.
- 10^{-16}
- , as a fraction

$$\frac{1}{10,000,000,000,000,000}$$

- b.
- $\left(\frac{1}{10^{-5}}\right)^3$

$$(10^5)^3 = 10^{15} = 1,000,000,000,000,000$$

- c.
- $10^{-9} \cdot 10^6$

$$10^{-3} = 0.001$$

8. Two measurements are 7.2×10^{-4} and 0.00000018.

a. Write 7.2×10^{-4} in standard form.

0.00072

b. Write 0.00000018 in scientific notation.

1.8×10^{-7}

c. Which of the two is greater? Explain how you know.

7.2×10^{-4} — -4 is greater than -7 , so its power of 10 is larger; $0.00072 > 0.00000018$.

9. Complete each comparison.

a. How many times larger is 8.4 than 2.1?

4

b. How many times larger is 10^{-3} than 10^{-7} ?

$10^4 = 10,000$

c. How many times larger is 8.4×10^{-3} than 2.1×10^{-7} ?

$4 \times 10^4 = 40,000$ times larger

10. Write each value in scientific notation.

a. 112 ten-thousandths

$0.0112 = 1.12 \times 10^{-2}$

b. 0.00112

1.12×10^{-3}

c. 112 billionths

$0.000000112 = 1.12 \times 10^{-7}$

11. Which is 0.00067 written in scientific notation?

☐ A. 67×10^{-5}

☐ B. 6.7×10^{-3}

☒ C. 6.7×10^{-4}

☐ D. 6.7×10^4

C

12. Select ALL of the expressions that are equal in value to 0.0034.

- ☒ A. 0.34×10^{-2}
- ☐ B. 3.4×10^3
- ☒ C. 3.4×10^{-3}
- ☒ D. 34×10^{-4}
- ☐ E. 3.4×10^{-2}
- ☒ F. 340×10^{-5}

A, C, D and F

Only C is in scientific notation.

13. A grain of pollen has a mass of about 8×10^{-9} gram. A bag of flour has a mass of about 2×10^3 grams.

a. How many times larger is the mass of the bag of flour? Write the answer in scientific notation.

2.5×10^{11} **times**

$2 \div 8 = 0.25$ and $3 - (-9) = 12$; the product $0.25 \text{ times } 10^{12}$ is rewritten with a coefficient in $[1, 10)$.

b. How many times smaller is the grain of pollen than the bag of flour?

2.5×10^{11} **times smaller — the same comparison read the other way.**